BILAL SIDDIQUI

Applied ML & Product Engineer

Islamabad, Pakistan | bilal.siddiqui@example.com | Professional experience: 4 years 5 months

PROFESSIONAL PROFILE

Applied ML and full-stack product engineer with 4 years 5 months of professional AI/ML delivery. Builds NLP/LLM products as testable software, with attention to dataset leakage, reproducibility, latency, and product telemetry.

TECHNICAL SKILLS

Python, PyTorch, transformers, scikit-learn, TensorFlow fundamentals	Machine learning, deep learning, named-entity recognition, information retrieval, RAG
LangChain, LangGraph, hosted LLM API integrations, tool-based agents, function calling	Accuracy, precision/recall, macro F1, BLEU/ROUGE limitations, calibration, retrieval recall
Google Cloud, Cloud Run, Vertex AI, BigQuery, Pub/Sub; Docker; GitHub Actions	Node.js/TypeScript APIs; Firebase Auth and Firestore; FastAPI

PROFESSIONAL EXPERIENCE

AI Product Engineer | CivicWave Software (fictional) | **Jul 2023 - Present**

- Architected an LLM-enabled document review service using Python, LangChain, LangGraph, retrieval, and human approval checkpoints.
- Built a Node.js web backend and Firebase Auth/Firestore front end for case reviewers and secure role-specific access.
- Developed PyTorch sentence classification and NLP entity-resolution models for multilingual complaint records.
- Prepared held-out evaluation sets; tracked retrieval recall@5, response faithfulness, false-positive extraction, and per-run cost.
- Deployed containers to Google Cloud Run and used Vertex AI for managed evaluations and release candidate comparisons.
- Implemented tracing with source IDs and reviewer overrides, reducing time spent reproducing reported response errors.
- Maintained staged rollback paths, versioned prompts, smoke tests and monthly prompt-injection exercises.
- Partnered with product managers on acceptance tests that separated tool reliability, LLM output quality and user-interface behavior.
- Documented PII handling, access permissions and model-provider configuration for operational audits.
- Introduced a response caching and batch-job queue that reduced recurring inference work while preserving per-user authorization.
- Reviewed integration tests comparing the original source segments with cited excerpts returned by the assistant.

Junior Machine Learning Engineer | Peak Data Labs (fictional) | **May 2022 - Jun 2023**

- Built Python ML pipelines for customer segmentation and ticket classification; documented data splits and baseline performance.
- Trained a PyTorch CNN for receipt-image category detection, measuring per-class recall and conducting ablation experiments.
- Collaborated on a forecasting service using rolling-origin validation and documented performance by seasonal segment.
- Published REST inference endpoints, Docker images, and simple monitoring metrics for internal consumers.

ENGINEERING DEPTH AND IMPACT

ENGINEERING PRACTICES

- Created structured evaluation releases that compare retrieval-source correctness, tool-call reliability, answer relevance and human-reviewed error types.
- Built production telemetry dashboards separating model request latency, data-access errors, retrieval performance and token spend by feature.
- Maintained staged CI/CD releases, secure Firebase user authentication, Cloud Run service identities and rollback procedures.
- Facilitated internal threat-model sessions on prompt injection, document exfiltration, unauthorized tool actions and source provenance.

ADDITIONAL ACCOMPLISHMENTS

- Refactored a classification pipeline into reusable Python training, inference and reporting modules with deterministic random seeds.
- Conducted weekly failure-review sessions with data annotators and domain reviewers to improve low-resource language performance.
- Designed PyTorch dataset loaders with class-balanced batches and reproducible evaluation notebooks for paper-trail audits.
- Integrated node-based user interface with signed backend requests, job-level progress tracking and reviewable confidence summaries.
- Published internal system diagrams explaining the difference between dense retrieval scores and task-level competence evidence.

RELATED PRODUCTION SKILLS

- Deployments: Google Cloud Run, Vertex AI experiments, Docker, GitHub Actions and PostgreSQL migrations.
- ML process: feature and data contracts, qualitative review, regression tracking, baseline comparisons and out-of-time evaluation.
- LLM applications: LangChain, LangGraph, retrieval chains, structured extraction and reviewable citations.

SELECTED PROJECT OUTCOMES

Retrieval-based case assistant: Held-out groundedness evaluation, LangGraph review transitions and per-document citation checks.

Customer service classifier: PyTorch; macro F1 0.87 in offline evaluation, reviewer spot-checks and error breakdown by language.

Operational forecasts: Time-series backtesting with seasonal baselines and weekly drift/error reporting.

EDUCATION, METHODS AND ADDITIONAL BACKGROUND

- BS Software Engineering | Metropolitan Institute (fictional), 2022
- Production ownership: alerting, rate limiting, caching, CI/CD, backup and release reviews
- Technical mentorship: pairing sessions on privacy, source lineage and API contracts
- Research practices: explicit failure cases, confidence intervals where applicable, reproducible seeds